5300 Final Presentation

FetchFood Penalty Analysis

Savannah T, Stefan I, Elling P

FetchFoods

While other delivery platforms allow drivers to cancel without any penalty, frequent cancellations by delivery drivers result in reduced profits for FetchFoods and its grocery and restaurant partners.

Goals

Reduce Cancellations

Our primary goal is to dissuade drivers from cancelling their deliveries by imposing a penalty fee.

Increase Profit

Our secondary goal is to increase our profit for the company and their grocery and restaurant providers

Customer Satisfaction

Cancelled orders can upset customers so our goal is to limit cancellations as much as possible

Metrics

Primary Metric

Cancel Rate

Cancellation rate per driver decreases as the penalty increases

Secondary Metrics

Revenue

Total revenue per driver increases as the penalty increases

Cancellation Impact on Revenue

The amount lost due to cancellations decreased in each treatment group

Guardrail Metrics

Requests

The number of requests recorded for a given driver during the study period

Completed Requests

The number of requests a given driver completes during the study

Data Cleaning/Preparation

- ➢ Removed NaN values
- ➢ Reformatted the Date
- ➢ Grouped by Driver
- ➢ Segmented by Business Type

Results & Analysis

Cancellation Rate and Revenue

- ➢ Cancellation Rate
- ➢ Revenue
- ➢ Profit Lost due to Cancellation

For these metrics we saw statistically significant differences between all penalty groups

Watching Driver Engagement

Requests

Fewer requests per driver suggests that drivers may be quitting sooner or becoming disengaged.

Completed Requests

Fewer completed orders per driver could mean losing out on potential revenue and failing customer expectations.

Key Point: If applied to all drivers, a 95% confidence lower bound of just 3.73% fewer requests per driver, as seen in the control vs the $20 penalty group, could result in missing out on revenue on the order $70000 during a six week period.

Results for Requests and Completed Requests

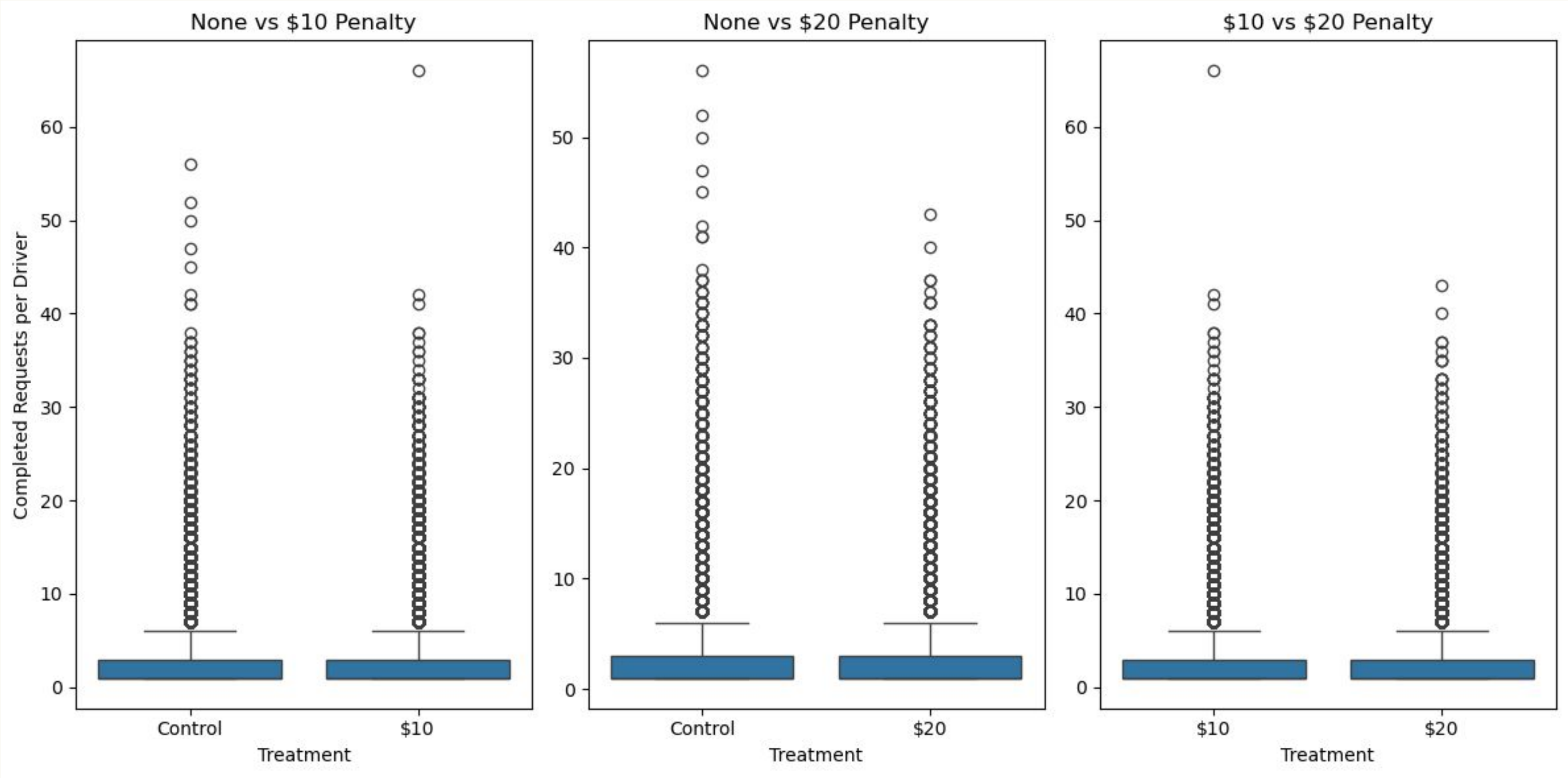
Requests:

- ➢ The decrease was significant from the control and $10 penalty groups to the $20 penalty group
- ➢ The difference from control to the $20 group was ~0.0542 orders per driver

Completed Requests:

- ➢ The decrease was significant from the control and $10 penalty groups to the $20 penalty group
- ➢ There was an increase from the control to the $10 penalty group
- ➢ The difference from control to the $20 group was ~0.0432 orders per driver

The decrease in completed requests is highly significant from other groups to the $20 penalty group. However, the effect is small and influenced by potential outliers.



Grocery vs Restaurant

Repeated primary and secondary metric tests between restaurant and grocery.

Goal:

1. To determine if the penalty fee impacts one business type more or less.
2. To determine if we could differently apply the penalty to a specific segment to make it be more effective.

Grocery vs Restaurant

Results

1. Cancellation rate:
   - Restaurant deliveries are being cancelled more than grocery deliveries.
   - Grocery segment is more sensitive to the penalty fee.
2. Total profit per driver:
   - All groups are significantly different.
   - Profit increases for both segments, but not equally.
3. Total profit lost from the penalty:
   - All groups are significantly different.
   - Profit is gained, not lost.

Hourly Distribution of Treatment Effect

Why?

1. The Restaurant segment was not as impacted by the cancellation fee.
2. If there are peak cancellation periods to determine if the penalty fee is less effective during certain times of day.

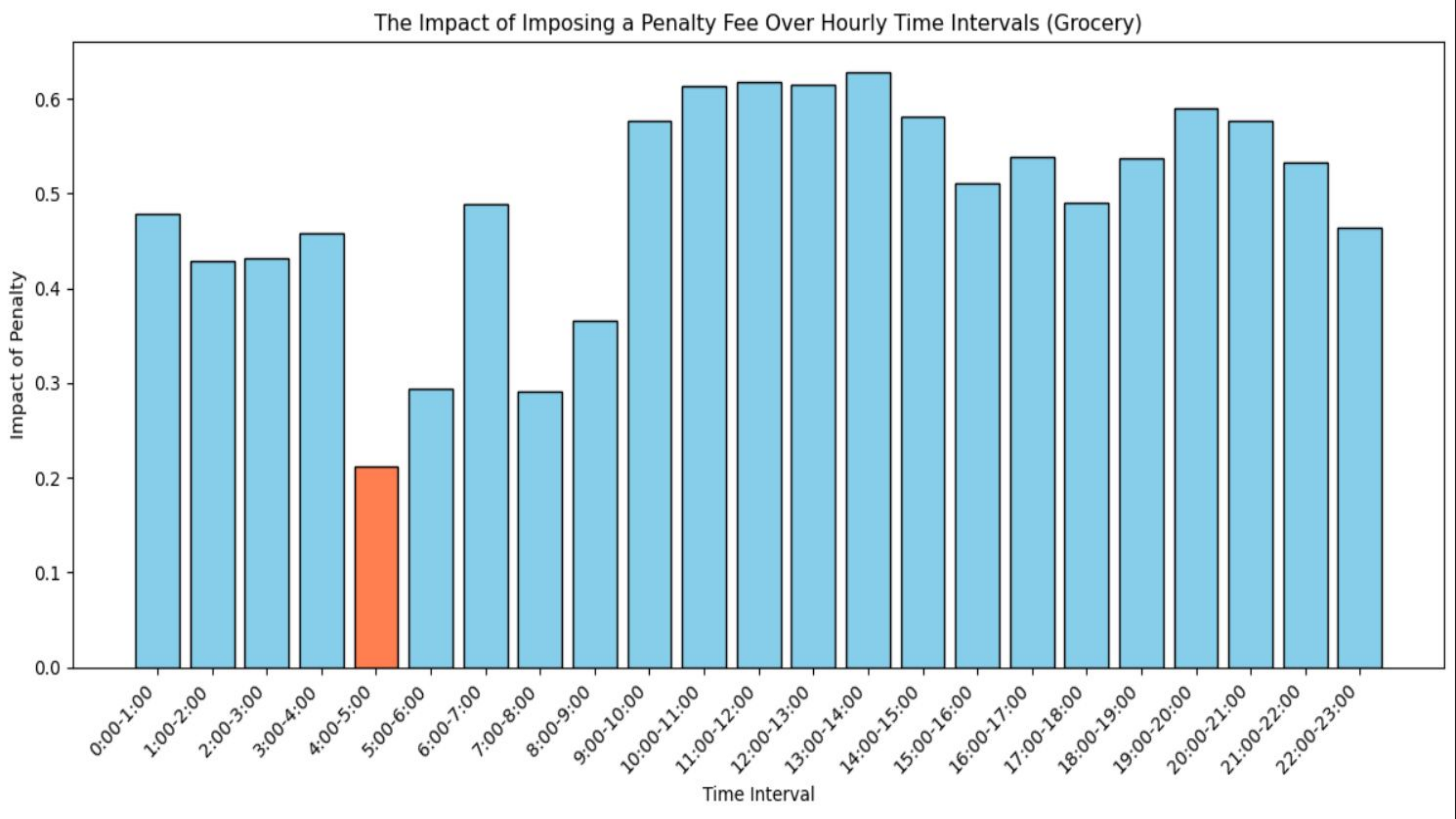
Goal:

1. Determine if the Restaurant segment have certain periods where there are more cancellations than normal.
2. Are there more selective ways to apply the fee during the day.

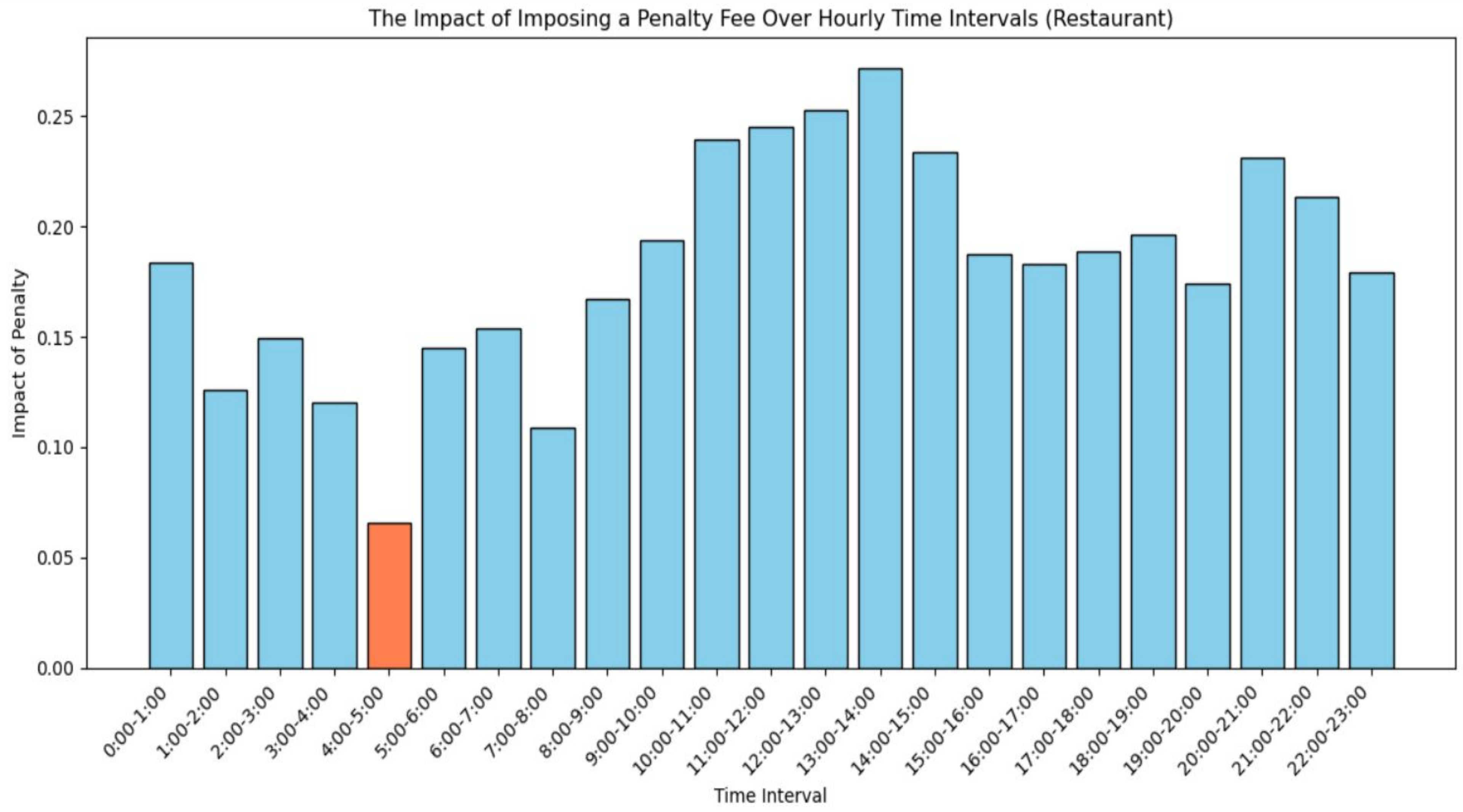
Hourly Distribution of Treatment Effect

<u>Methods:</u>

1. Calculated total orders cancelled per hourly interval for all groups, including segments.
2. Used chi squared tests for homogeneity to determine if the distributions across the control and treatment groups are equal or not.
3. Calculated the treatment effect of the penalty fee on the significant groups.
4. Calculated the overall ratio of cancellations per hourly interval.



The Impact of Imposing a Penalty Fee Over Hourly Time Intervals (Grocery)
Impact of Penalty
Time Interval
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0:00-1:00
1:00-2:00
2:00-3:00
3:00-4:00
4:00-5:00
5:00-6:00
6:00-7:00
7:00-8:00
8:00-9:00
9:00-10:00
10:00-11:00
11:00-12:00
12:00-13:00
13:00-14:00
14:00-15:00
15:00-16:00
16:00-17:00
17:00-18:00
18:00-19:00
19:00-20:00
20:00-21:00
21:00-22:00
22:00-23:00


The Impact of Imposing a Penalty Fee Over Hourly Time Intervals (Restaurant)
Impact of Penalty
0.00
0.05
0.10
0.15
0.20
0.25
0:00-1:00
1:00-2:00
2:00-3:00
3:00-4:00
4:00-5:00
5:00-6:00
6:00-7:00
7:00-8:00
8:00-9:00
9:00-10:00
10:00-11:00
11:00-12:00
12:00-13:00
13:00-14:00
14:00-15:00
15:00-16:00
16:00-17:00
17:00-18:00
18:00-19:00
19:00-20:00
20:00-21:00
21:00-22:00
22:00-23:00
Time Interval

Hourly Distribution of Treatment Effect

<u>Results:</u>

1. The distribution between Restaurant and Grocery for all penalty groups are significantly different.
2. The distribution between control and treatment groups for Grocery are significantly different.
3. The distribution between control and treatment groups for Restaurant are *not* significantly different.
4. The penalty fee has the *least* impact between 4 and 5 am for all deliveries.
5. Upon further inspection, the fee only has the *least* impact between 4 and 5 am for *Grocery* deliveries.

Conclusion and Recommendations

Penalties seem to increase revenue and decrease cancellations but may increase driver churn and disengagement, to which FetchFoods is vulnerable.

A $10 penalty appeared to improve cancellations and revenue without appearing to affect driver engagement metrics so far.

It is recommended that the company continue to cautiously explore lower penalty fees of $10 or less.

More Suggestions for Future Experiments

Grocery & Restaurant analysis

Data collection for Driver Churn

More Selective Penalties